Functional Description

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1. Process Operation

The process is used to prepare a mixture from water, powder and sodium hydroxide. The materials are added to the Mix Tank, mixed and then transferred to the Storage Tank. From the Storage Tank, the mixture can be sent to the next process or circulated back to the tank.

The Mix Tank has a capacity of **2 m^3** and the Storage Tank has a capacity of **5 m^3**.

Water supply

The process starts when valve **HS 510** opens. Water flows through the **DN25** pipe into the Mix Tank.

The weight of the material is measured by **WI 512**. When the required water weight is reached, HS 510 closes.

Powder and mixing

After the water is added, the agitator motor **M002** starts. M002 is a **0.75 kW** motor and mixes the material inside the tank.

The powder feeder motor **M001** also starts. M001 is a **4 KW** motor and brings powder from the hopper into the Mix Tank.

The temperature is monitored by **TI 517**. When the required amount of powder has been added, M001 stops. M002 continues mixing the material.

Sodium hydroxide

After the powder has been added and mixed, valve **HS 511** opens.

Sodium hydroxide flows through the **DN15** pipe into the Mix Tank. WI 512 measures the total weight in the tank.

When the final weight is reached, HS 511 closes.

Transfer to Storage Tank

When mixing is finished and there is enough space in the Storage Tank, **HS 514** opens.

The mixture flows through the **DN50** pipe into the Storage Tank.

The Storage Tank level is measured by **LI 515**. If the tank becomes full, HS 514 closes.

Delivery and recirculation

When the mixture is needed by the next process, pump **M003** starts. M003 is a **2.2 kW** pump.

The mixture goes through the **DN32** pipe. **FIC 516** controls the flow by changing the speed of M003 through a frequency converter.

There are two routes:

To process:

- HS 519 is open.
- HS 521 is closed.

Recirculation:

- HS 519 is closed.
- HS 521 is open.

In the recirculation mode, the mixture goes through the **DN25** pipe and returns to the Storage Tank.

The manual valve at the bottom of the Storage Tank is only used for cleaning and maintenance. It is not part of the automatic control system.

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2. Functional Description

2.1 Mix Tank Dosing and Mixing

The first control function is the dosing and mixing in the Mix Tank.

Measured:

- WI 512 – tank weight
- TI 517 – temperature

Controlled:

- HS 510 – water valve
- M001 – powder feeder
- M002 – agitator
- HS 511 – sodium hydroxide valve

First, HS 510 opens and water enters the tank. WI 512 measures the weight. When the water target is reached, HS 510 closes.

M002 and M001 then start. M002 mixes the material and M001 feeds powder into the tank.

When the required powder amount is reached, M001 stops. M002 continues mixing.

After the powder mixing stage, HS 511 opens and sodium hydroxide is added. When the final batch weight is reached, HS 511 closes.

TI 517 monitors the temperature during the process.

2.2 Mix Tank Transfer

The second control function is transferring the mixture from the Mix Tank to the Storage Tank.

The main measurements are **WI 512**, **LI 515** and **TI 517**.

The controlled valve is **HS 514**.

HS 514 opens when the mixing is finished and there is enough free space in the Storage Tank.

The mixture then flows through the DN50 pipe into the Storage Tank.

If the Storage Tank reaches its maximum level, HS 514 closes.

HS 514 also closes when the Mix Tank reaches its minimum weight.

2.3 Storage Tank Level

The Storage Tank level is measured by **LI 515**. The temperature is monitored by **TI 518**.

LI 515 is used to monitor the amount of mixture in the tank.

If the level becomes too high, HS 514 closes to prevent overflow.

If the level becomes too low while M003 is running, M003 stops. This prevents the pump from running without enough liquid.

2.4 Pump and Flow Control

The final control function is transferring the mixture from the Storage Tank.

When the mixture is needed by the next process, **HS 519 opens and M003 starts**.

The flow is controlled by **FIC 516**. The controller changes the speed of M003 using the frequency converter to keep the required flow.

If the mixture is not needed, it must be recirculated.

In this case, **HS 519 is closed and HS 521 is open**.

HS 519 and HS 521 cannot be open at the same time.

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3. Alarms and Interlocks

The powder hopper has a low-level alarm **LSA 513**. This informs the operator when the powder level is low.

If the Storage Tank reaches its high-level limit, **HS 514 closes** to prevent overflow.

If the Storage Tank level becomes too low during pumping, **M003 stops** to protect the pump.

If the Mix Tank temperature becomes too high, the process is stopped or prevented from continuing.

If an automatic valve does not reach its required position, a valve fault is generated, and the related process should not continue.

During an emergency stop:

- M001 stops.
- M002 stops.
- M003 stops.
- HS 510 closes.
- HS 511 closes.
- HS 514 closes.

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4. Startup and Shutdown

Before starting, the system checks that the emergency stop is reset, sensors are working and there are no active interlocks.

The main valves are closed, and the motors are stopped before the automatic sequence starts.

During normal shutdown, the dosing valves close and the motors and pump are stopped in a controlled way.

During an emergency shutdown, the motors stop and the automatic valves close.

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5. Process Sequence

The main sequence is:

Start → Water dosing → Powder dosing and mixing → Sodium hydroxide dosing → Mixing complete → Transfer to Storage Tank → Process delivery or recirculation

The control system uses the sensor signals to decide when each step is completed before moving to the next step.

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6. Artificial Intelligence Usage

I used **ChatGPT and Gemini** while preparing this functional description.

I used AI mainly to help me understand how to write the functional description and how to divide the process into different control loops.

I used AI especially for the **Mix Tank Dosing and Mixing Control Loop**. It helped me organise the information about measurements, controlled equipment, sequence, alarms and interlocks.

I did not use the AI text directly. I compared the suggestions with the P&I diagram and changed the information where necessary.

I checked the component tags, pipe sizes, motor powers, tank capacities and process sequence against the P&I diagram.

The final responsibility for the technical content was mine. I had to understand the process, check the information and decide what should be included in the final document.

AI helped me understand that a functional description should explain what is measured, what is controlled, how the control works and what happens during faults and shutdown.

Important prompts

- "Can you rewrite this with improvement in very simple language?"
- "Rewrite Software Functional Description in very simple language for the dissolving unit control loops."
- "For this Introduction and production overview how, much point can you give out of 10?"
- "Assuming I am appointed your assignment teacher, how much grade can you give me out of 5?"

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7. Conclusion

This functional description explains the main operation of the mixing and storage process.

Water, powder and sodium hydroxide are added to the Mix Tank and mixed together. The mixture is then transferred to the Storage Tank.

From the Storage Tank, M003 sends the mixture to the next process or back through the recirculation line.

The sensors, valves, motors and interlocks help the process work correctly and safely.